

MathMind Application

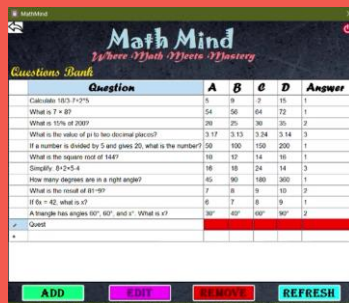
App features

This Multiple Choice Online Exam System provides several special features that facilitate the experience for both examiners and students.

In case of examiners: there is a question bank stored in database that can be accessed by examiners, in this question bank repository, examiners can add or delete or edit question and their answers.

In case of students: they send a request to examiner which may be accepted or rejected by the examiner. Once the request is approved, the quiz will be sent to the student and a two minutes timer will start.

Examiners and students should be registered in the system in order to request quizzes (students) or access the questions bank (examiners), resulting in enhancing the security and privacy.



Problem statement

Nowadays, examiners and students face multiple struggles in managing and accessing quizzes. Therefore, we have created a user-friendly app that eliminates the gap between students and examiners, enabling seamless quiz creation and manipulation and also feedback to boost the learning experience.

Proposed model

Our app ready solution allows students to request quizzes and exams from their tutors (examiners), solve the quiz and submit the answers and retrieve their tutor's feedback. Also, it allows the examiners to easily track their students' performance throughout the year using this game-like app (which contributes in intriguing the students to learning) and constantly help the students improve their skills.

Proposed by

- Mohamed Eldairouty
- Jomana Emad
- Malak Amro
- Ziad Magdy
- Omar Yakout

As part of ECE4301 Computer Networks course.

Submitted to: Dr. Mohamed Elserafy, Eng. Dannah Khedr, Eng. Marawan Nour

Servers(examiners)

Examiners act as servers in this system, serving dynamic quiz content to students. They have privileged access to the database, allowing them to retrieve, edit and send quiz content.

- The system follows a client-server architecture which was used by .Net package and C# language (using the concepts of networking such as TCP). A text file that serves as a log file for tracking interactions, a configuration storing sensitive settings like database credentials or IP addresses for connection

Clients(students)

Students act as clients in the system, initiating requests for quiz content.

- Note that the database is dynamic and can be accessed by both examiners and students. All types of exception handling have taken place in this app to increase efficiency and improve security.